Real-Time Schedulers for STM32: Performance Evaluation of EDF over Superloop

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I. Introduction

STM32 is a family of microcontrollers developed by STMicroelectronics for embedded applications. These devices integrate a processor, memory, and peripheral interfaces on a single chip [1].

STM32-based devices support applications such as environmental monitoring [2] and equipment condition monitoring [3]. In these applications maintaining responsiveness as the workload increases is a key challenge.

The paper examines how the choice of scheduling approach affects responsiveness and processing cost on STM32. We compare two approaches: Superloop [4] and Earliest Deadline First (EDF) [5].

A. Related works

- Superloop in RTS.
- EDF and chunked EDF.
- Our evaluation.

B. Contribution

First, we compare Superloop and chunked EDF for periodic tasks with implicit deadlines on an STM32F767 microcontroller. Then, we evaluate the effects of task-set utilization on response time, deadline misses, skipped releases, and measured task-selection overhead. Also, we investigate how maximum chunk duration affects these metrics for chunked EDF. Both scheduler implementations are publicly available.

C. Paper organization

Initially, in Section II, we provide the necessary background on STM32. We describe the hardware platform in Section II-A and the project structure in Section II-B. Sections II-C and II-D explain how application functions execute and introduce the computation and sensor tasks used in our application. Next, in Section III we formalize the periodic task model used in our experiments. Finally, in Section IV, we describe Superloop, fully preemptive EDF, and chunked EDF and illustrate their schedules for the same task set.

II. Background on STM32

A. Hardware platform

We use an STM32F767 microcontroller with four sensors in our experiments (Fig. 1). The sensors collect different types of data and use different interfaces, as shown in Table I. The code that reads and processes their data forms the sensor-based workload for our scheduling experiments.

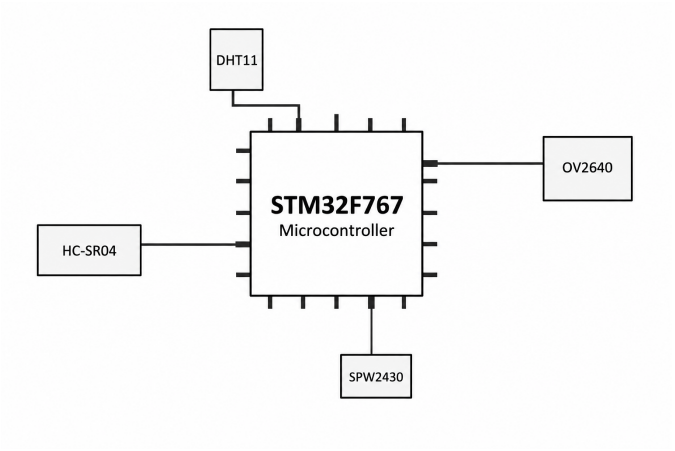


Fig. 1. Sensor platform based on STM32F767.

TABLE I
Sensor functions and interfaces.

Sensor	Purpose	Interface
DHT11 [6]	Temperature and humidity	Single-wire digital, GPIO
HC-SR04 [7]	Distance	Trigger/echo, GPIO
OV2640 [8]	Image acquisition	Parallel data, SCCB
SPW2430 [9]	Signal acquisition	Analog output, ADC

GPIO - General-Purpose Input/Output; SCCB - Serial Camera Control Bus; ADC - Analog-to-Digital Converter.

B. STM32 Project Structure

We use STM32CubeMX to configure the microcontroller pins and peripherals and generate a C project for STM32CubeIDE where we can debug, execute and edit the code. Fig. 2 shows the main project components. Application source files and headers are placed in `Core/Src` and `Core/Inc`, respectively. The source files include `main.c`, which contains system initialization and the main loop from which application functions are called.

C. Default execution model

The generated `main()` function first initializes the system and peripherals, then enters an infinite `while (1)` loop. (Listing 1). STM32CubeMX marks `USER CODE` regions where developers can add their own code without losing it when the project is generated again. By default, the loop contains no application functions.

```
example_project/
├─ Core/ ..... Application code
│  ├─ Inc/ ..... Header files
│  └─ Src/ ..... Source files
│     ├─ main.c ..... Init. and main loop
│     ├─ stm32f7xx_it.c ..... Interrupt handlers
│     ├─ stm32f7xx_hal_msp.c ..... Low-level peripheral setup
│     └─ system_stm32f7xx.c ..... Core init. and clock state
│  └─ Startup/ ..... Startup code and interrupt vectors
├─ Drivers/ ..... Hardware support
│  └─ CMSIS/ ..... Core and MCU definitions
├─ STM32F767ZITX_FLASH.ld
├─ STM32F767ZITX_RAM.ld
└─ test_project.ioc ..... CubeMX configuration
```

Fig. 2. Main components of the STM32CubeMX-generated project.

Listing 1. Simplified STM32CubeMX-generated main function showing system initialization, user-code regions, and the infinite loop.

```
int main(void) {
  /* CubeMX-generated initialization */
  MPU_Config();
  HAL_Init();
  SystemClock_Config();
  MX_GPIO_Init();
  /* Other peripheral initialization */

  /* USER CODE BEGIN 2 */
  /* User-defined application setup */
  /* USER CODE END 2 */

  /* USER CODE BEGIN WHILE */
  while (1)
  {
    /* USER CODE END WHILE */

    /* USER CODE BEGIN 3 */
    /* User-defined functions */
  }
  /* USER CODE END 3 */
}
```

Application functions can be called from inside this loop. In a Superloop, these functions run in a fixed order, and the sequence repeats after the last function returns [4] (Fig. 3). For example, `taskB()` runs only after `taskA()` returns, so the execution time of one function affects when the next function can start.

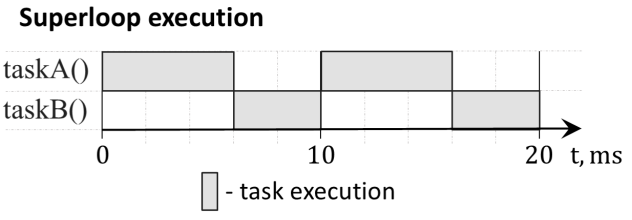


Fig. 3. Example of Superloop execution. Functions are executed sequentially in a fixed order, and the loop repeats indefinitely.

D. User defined tasks

1) Synthetic tasks: We use synthetic tasks to create CPU workload without reading data from sensors. Listing 2 shows one example, where a function updates a stored value through 10,000 iterations. The number of iterations can be changed to control the amount of computation, although the actual execution time must be measured on the STM32.

Listing 2. Example of a synthetic computation.

```
static volatile uint32_t state = 1U;

void SyntheticTask_Run(void)
{
  uint32_t value = state;

  for (uint32_t i = 0; i < 10000U; i++) {
    value = value * 1664525U + i;
  }

  state = value;
}
```

2) Sensor tasks: Unlike synthetic tasks, sensor tasks must communicate with external devices. Listing 3 shows how the STM32 reads data from a DHT11 sensor. The function first holds the data pin low for 30 ms and then reads the sensor response. With this blocking implementation, the main loop waits for the sensor read to complete before calling another function.

Listing 3. Example of a DHT11 sensor read operation.

```
int DHT11_Read(uint8_t *temp, uint8_t *hum)
{
  DHT11_SetOutput();

  HAL_GPIO_WritePin(DHT11_PORT,
  DHT11_PIN, GPIO_PIN_RESET);

  HAL_Delay(30);

  return DHT11_ReadTransaction(temp, hum);
}
```

Sensor readings also have timing limits. For example, the DHT11 requires at least 1 second between readings [6], while the HC-SR04 recommends more than 60 ms between measurements [7]. Calling these functions on every pass through the main loop may violate these limits.

III. System Model

We use the periodic task model of Liu and Layland [5] and adapt the task and release notation of Burmyakov and Bini [11]. Table II summarizes the notation.

A real-time system $\mathcal{T} = \{\tau_1, \ldots, \tau_n\}$ consists of n periodic tasks executed on one core ($m = 1$). Each task $\tau_i = (C_i, D_i, P_i)$ is defined by the worst-case execution time C_i of a job, a relative deadline D_i, and a period P_i. We consider implicit deadlines, $D_i = P_i$.

Let $J_{i,j}$ denote the j-th job of τ_i, released at $r_{i,j}$. Consecutive jobs are released one period apart:

$$r_{i,j+1} = r_{i,j} + P_i. \tag{1}$$

Each job $J_{i,j}$ has an absolute deadline $d_{i,j} = r_{i,j} + D_i$. If completed, its finish time is denoted by $f_{i,j}$.

TABLE II
SYSTEM MODEL NOTATION.

Symbol	Meaning
$\mathcal{T}, n$	Task set and number of tasks
τ_i	Task i
C_i	Worst-case execution time of a job of τ_i
P_i	Task period
$D_i = P_i$	Relative deadline, equal to the period
$J_{i,j}$	The j-th job of task τ_i
$r_{i,j}$	Release time of job $J_{i,j}$
$f_{i,j}$	Job completion time
$d_{i,j} = r_{i,j} + D_i$	Absolute deadline of job $J_{i,j}$
m	Number of cores
$U_i = C_i/P_i$	Task utilization
$U_\mathcal{T} = \sum_{\tau_i \in \mathcal{T}} U_i$	Task-set utilization

- A schedule specifies which job executes at each time, if any.
- Preemption interrupts a job before completion to execute another job. The interrupted job may resume later.
- A completed job $J_{i,j}$ misses its deadline if $f_{i,j} > d_{i,j}$.
- The response time of a completed job $J_{i,j}$ is $f_{i,j} - r_{i,j}$.

IV. SUPERLOOP AND EDF SCHEDULERS

We next describe the Superloop and chunked EDF schedulers used in our experiments. Fully preemptive EDF serves as a reference for comparison.

Both implementations use the same policy for missed releases. After a job completes, its next release is advanced by one period. Any subsequent releases earlier than its completion time are skipped. This avoids catch-up executions that may be unnecessary for monitoring tasks.

A. Superloop

The key features of the Superloop scheduler:

- periodically invoking;
- non-preemptive;
-

since it determines which tasks to execute based on their release times and a fixed order in the main loop.

At the start of each loop iteration, the scheduler checks the readiness of all tasks. It then executes the ready tasks in a fixed order. Each job runs to completion before the next one starts. Similar approaches to periodic scheduling in a main loop are described by Koopman [10] and Jones [12]. Algorithm 1 shows this process.

Algorithm 1 Superloop Scheduler

```
1:  function SUPERLOOP(T)
2:      t_i^next ← 0,   i = 1, ..., n          ▷ Initial releases
3:      while true do                           ▷ Infinite cycle
4:          t^cur ← NOW()                       ▷ Get current system time
5:          for i ← 1 to n do
6:              b_i ← (t_i^next ≤ t^cur)        ▷ Save readiness
7:          end for
8:          for i ← 1 to n do
9:              if b_i then
10:                 J_{i,j} ← (τ_i, t_i^next)   ▷ Keep release time
11:                 EXECUTE(J_{i,j})
12:                 f_{i,j} ← NOW()
13:                 t_i^next ← t_i^next + P_i   ▷ Advance, then skip
14:                 while t_i^next < f_{i,j} do t_i^next ← t_i^next + P_i
15:             end if
16:         end for
17:     end while
18: end function
```

Fig. 4(a) illustrates the Superloop schedule for tasks τ_1 and τ_2. Both tasks release their first jobs at $t = 0$. Since τ_1 comes first in the polling order, $J_{1,1}$ executes until $t = 18$ ms. Job $J_{2,1}$ then starts and completes at $f_{2,1} = 28$ ms, missing its deadline $d_{2,1} = 20$ ms.

The release at $r_{2,2} = 20$ ms occurs while $J_{2,1}$ is still running. Once this job completes, the scheduler counts that release as skipped and advances the next release time to $r_{2,3} = 40$ ms. Polling continues with no ready job until then.

B. Earliest Deadline First

In contrast, fully preemptive EDF selects the ready job with the earliest absolute deadline. A newly released job preempts the running job if its deadline is earlier.

Fig. 4(b) shows the EDF schedule for the same task set. At $t = 0$, $J_{2,1}$ has the earlier deadline and executes until $t = 10$ ms. Job $J_{1,1}$ then starts, but is preempted at $r_{2,2} = 20$ ms because the new job has a deadline $d_{2,2} = 40$ ms, earlier than $d_{1,1} = 50$ ms. Job $J_{2,2}$ executes until $t = 30$ ms, after which $J_{1,1}$ resumes and completes at $t = 38$ ms. The next job of τ_2 executes from $t = 40$ to $t = 50$ ms. Thus, no deadline is missed and no release is skipped in this example.

C. Chunked Earliest Deadline First

Unlike fully preemptive EDF, our chunked EDF implementation executes synthetic workloads in non-preemptive chunks of at most Δ ms.

Algorithm 2 outlines chunked EDF. The Boolean flag a_i indicates whether task τ_i has an unfinished job. At each iteration, $\mathcal{R}$ contains the indices of runnable tasks. These include tasks with due releases and active jobs that can resume. Sensor jobs waiting for hardware are excluded.

The scheduler selects the job with the earliest absolute deadline. For an active job, $d_i = r_i + D_i$, where r_i is its saved release time. Otherwise, $d_i = t_i^{\text{next}} + D_i$. The selection is repeated after each execution step.

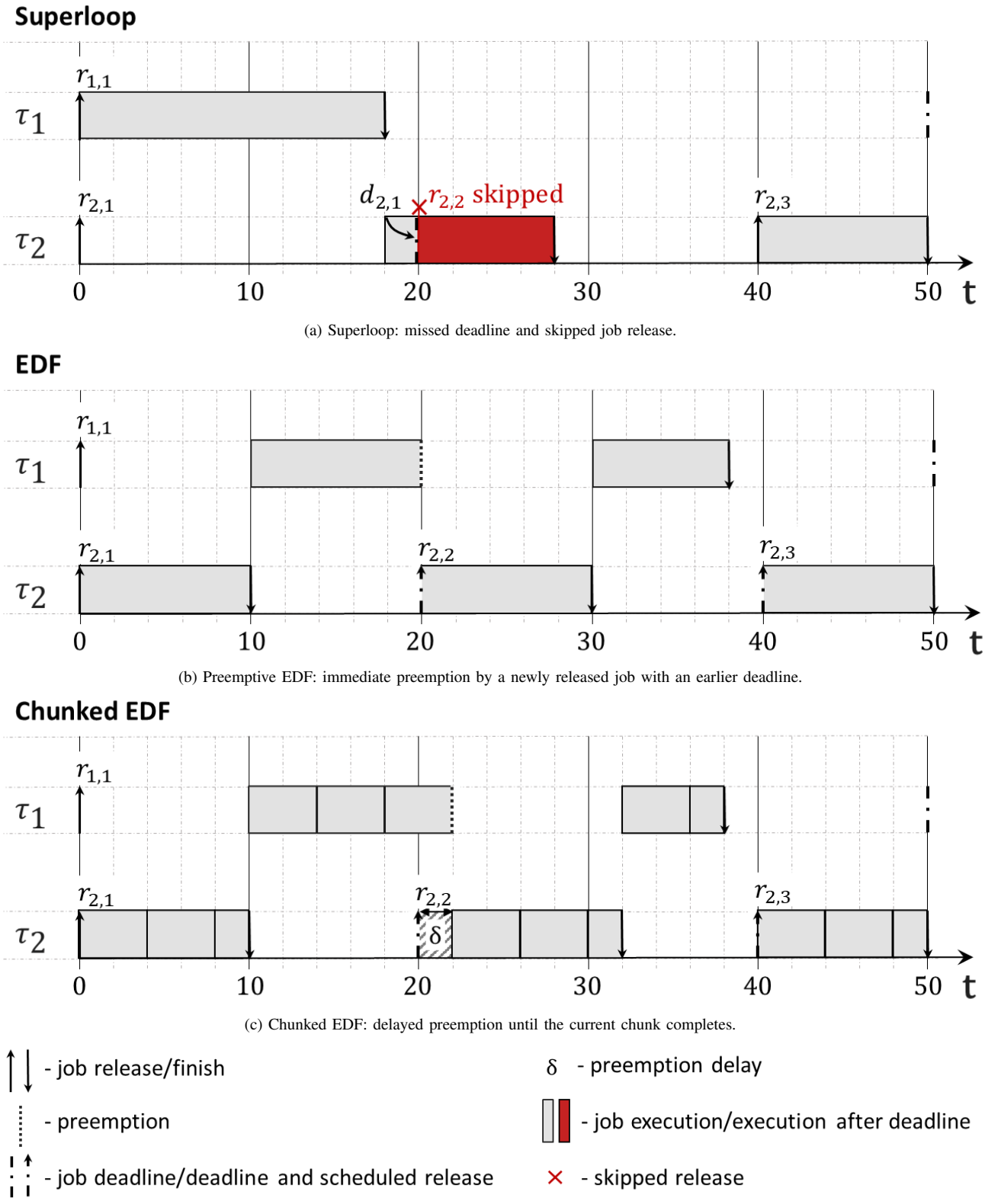
Fig. 4(c) shows the same task set with $\Delta = 4$ ms. At $r_{2,2} = 20$ ms, the first job of τ_1 is still executing a chunk, which ends at $t = 22$ ms. The new job of τ_2 therefore starts at $t = 22$ ms, with a preemption delay of $\delta = 2$ ms. It executes in chunks of 4, 4, and 2 ms and completes at $t = 32$ ms. The shorter final chunk reflects the remaining work. The interrupted job of τ_1 then resumes and completes at $t = 38$ ms.

Algorithm 2 Chunked EDF Scheduler

function CHUNKEDEDF($\mathcal{T}, \Delta$)
  $t_i^{\mathsf{next}} \leftarrow 0, \quad a_i \leftarrow \mathsf{false}, \quad i = 1, \ldots, n$
  while true **do** ▷ Infinite cycle
    $t^{\mathsf{cur}} \leftarrow$ NOW() ▷ Get current system time
    $\mathcal{R} \leftarrow \{i \mid \text{READY}(i, t^{\mathsf{cur}})\}$ ▷ Ready task indices
    if $\mathcal{R} \neq \emptyset$ **then**
      $i^\star \leftarrow \arg\min_{i \in \mathcal{R}} d_i$ ▷ Earliest deadline
      if $\neg a_{i^\star}$ **then**
        $r_{i^\star} \leftarrow t_{i^\star}^{\mathsf{next}}$; $a_{i^\star} \leftarrow \mathsf{true}$ ▷ Save release
      end if
      EXECUTESTEP($i^\star, \Delta$) ▷ At most Δ
      if COMPLETED($i^\star$) **then**
        $f \leftarrow$ NOW()
        $t_{i^\star}^{\mathsf{next}} \leftarrow t_{i^\star}^{\mathsf{next}} + P_{i^\star}$ ▷ Advance, then skip
        while $t_{i^\star}^{\mathsf{next}} < f$ **do** $t_{i^\star}^{\mathsf{next}} \leftarrow t_{i^\star}^{\mathsf{next}} + P_{i^\star}$
        $a_{i^\star} \leftarrow \mathsf{false}$
      end if
    end if
  end while
end function

$$\tau_i = (C_i, D_i, P_i), \qquad D_i = P_i, \qquad m = 1$$
$$\tau_1 = (18, 50, 50), \qquad \tau_2 = (10, 20, 20)$$

All times in ms. Maximum chunk duration in (c): $\Delta = 4$ ms.



(a) Superloop: missed deadline and skipped job release.

(b) Preemptive EDF: immediate preemption by a newly released job with an earlier deadline.

(c) Chunked EDF: delayed preemption until the current chunk completes.

Fig. 4. Comparison of Superloop, preemptive EDF, and chunked EDF schedules for the same periodic task set.

V. Performance evaluation

A. Synthetic workload

B. Real sensors workload

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